Beyond the hourly average

Quarter-hour electricity price reconstruction

Challenge proposed by **AET – Azienda Elettrica Ticinese**

The challenge

Our task was to reconstruct **96 quarter-hour electricity prices from 24 supplied hourly prices**, for each of four market zones. An hourly average hides movements between its four quarters. A more detailed curve could therefore help AET understand price differences within the hour and support valuation when electricity is delivered unevenly.

Using the official price data, we had to recover that detail while preserving each hourly average within **0.0001 €/MWh**. We aimed for a solution we could both evaluate and explain.

Approach: learn the shape, preserve the average

Our EDA highlighted slope: we combine a smooth baseline with learned corrections for sharper movements.

The prediction combines a smooth PCHIP baseline with a learned correction:

$$\widehat{P}_{h,q} = \underbrace{B_{h,q}}_{\text{smooth baseline}} + 0.8 \times \underbrace{R_{h,q}}_{\text{learned correction}} .$$

For hour h and quarter q, three LightGBM models learn the weights of the following shapes. Values correspond to **Q1, Q2, Q3, Q4**, in that order:

Basis	Pattern	Normalized values used by the model
Slope	$[-3, -1, 1, 3]$	$[-0.671, -0.224, 0.224, 0.671]$
Curvature	$[1, -1, -1, 1]$	$[0.5, -0.5, -0.5, 0.5]$
Zigzag	$[-1, 3, -3, 1]$	$[-0.224, 0.671, -0.671, 0.224]$

Table 1: The three within-hour patterns. Normalized values are rounded; the model uses full precision.

Each vector sums to zero, so any weighted combination does too. The learned coefficients change the within-hour prices without changing their average, even after applying the fixed scale of 0.8.

Inputs include the day's hourly curve, differences between zones, calendar information and earlier price patterns. The target day's actual quarter-hour prices are excluded. Training minimizes coefficient-level absolute error; selection uses price-level MAE. The fixed 80% correction scale improved development results.

Final adjustments. Mean-preserving rules align shapes for zones with nearly equal hourly prices and restrict within-hour sign changes. Together with increasing boosting rounds from 500 to 1,000, they reduced development MAE by 1.5% across all eight folds. These are empirical choices, not market rules.

Challenges and solutions

Missing information and fair validation. Many quarter-hour profiles share the same hourly average, so some detail remains unpredictable. We used chronological validation and calendar-based history lookups to avoid future information and incorrect lags when dates are missing.

Deciding when to stop adding complexity. We chose the final model because it offered a good balance between accuracy, simplicity and interpretability. Local-geometry features slightly worsened development MAE (3.9479 to 3.9497 €/MWh), so we retained the existing features. We also explored graph models, error-correction models and ensembles. Some achieved similar or slightly lower errors. However, a graph blend improved MAE by only 0.4% in a separate June–August comparison, too little to justify the added complexity. Separating the baseline, three corrections and rules makes predictions easier to explain in the dashboard and investigate, while allowing individual components to change without losing hourly consistency.

Results: more detail, less reconstruction error

We measure accuracy with mean absolute error (MAE): the average distance between reconstructed and observed prices. Each evaluation day and zone has equal weight; lower is better. The final model reduces MAE by **52.2% on development dates** and **54.1% on September**, compared with repeating hourly prices.

Development uses eight chronological January–August folds, training on earlier dates and scoring days 19 through month-end. September covers 19–30 September, with training through August. September was examined in several experiments before the final changes, so it is a **historical assessment, not an untouched final test**.

Method	Development MAE	September MAE
Flat hourly prices	8.13	10.90
PCHIP baseline	5.62	6.93
Full correction (scale 1.0)	3.98	5.43
Previous model (no rules)	3.95	5.04
Final model	**3.89**	**5.00**

Table 2: Average price error in €/MWh. All September scores use the same dates. The previous model uses 500 boosting rounds and no final adjustment rules; the final model uses 1,000 rounds and the rules.

Zone	Model MAE	Flat MAE	Error reduction
ZONE_0	3.83	10.78	64.5%
ZONE_1	6.12	13.30	54.0%
ZONE_2	5.49	9.72	43.5%
ZONE_3	4.57	9.79	53.3%

Table 3: September results by zone. Reductions compare the final model with flat hourly prices and are calculated before rounding. Source: `results/summary.csv`.

The model improves on flat prices in every zone. The final changes bring a smaller September gain over the previous model (0.8%), with ZONE_3 slightly worse (4.57 versus 4.55). These are price-accuracy results; effects on trading or portfolio valuations have not been measured.

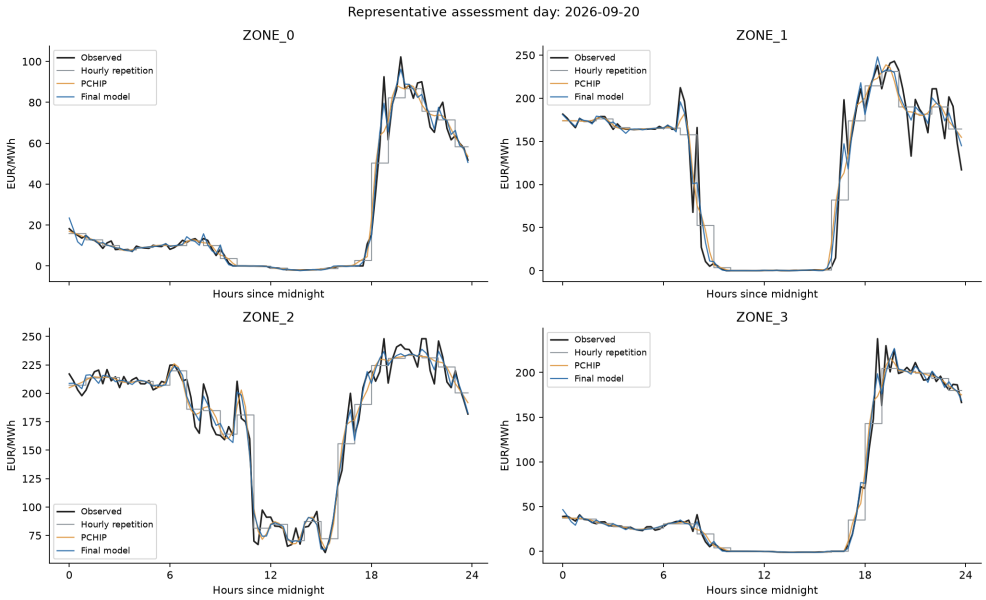


Figure 1: Reconstruction on 20 September 2026 across all four zones. The observed curve shows both where the model recovers detail and where sharp movements remain difficult.

Dashboard: making the predictions understandable

We also wanted someone without a machine-learning background to be able to explore the results. The dashboard brings together **Model Results**, **Understanding the Model**, **Risk Alerts** and the **Data Story**. Users can compare price curves, inspect one hour's component contributions and see the signal behind its risk flag.

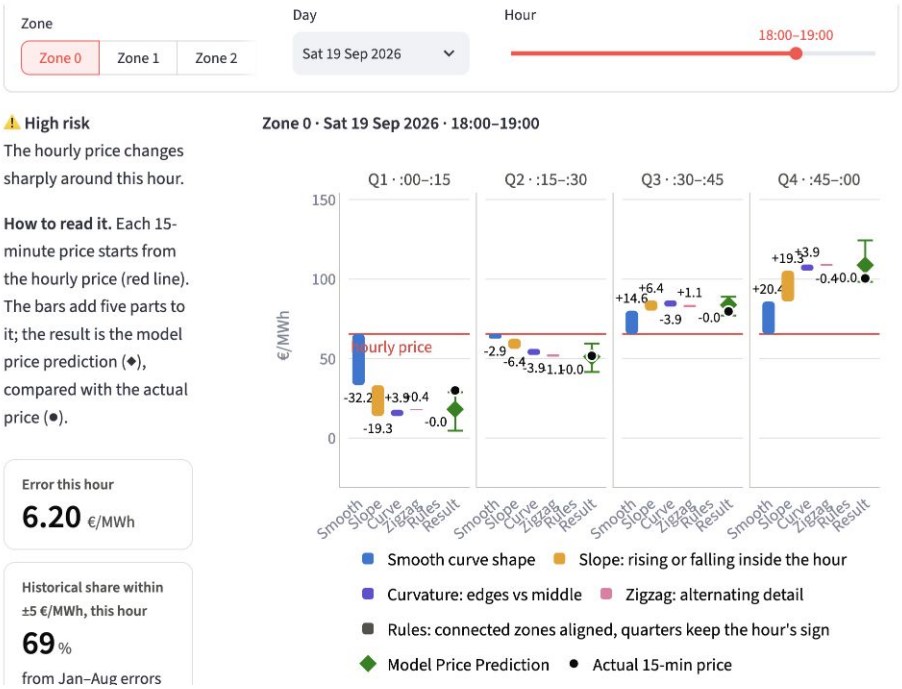


Figure 2: ZONE_0, 19 September 2026, 18:00–19:00. The dashboard separates the smooth curve, learned corrections and fixed rules, and compares the result with observed prices.

Does the risk indicator identify harder hours?

The indicator combines hourly ramps, daily variability and recent historical reconstruction difficulty. On development folds, high-risk hours had **2.8–3.9 times the MAE** of low-risk hours. This suggests that the labels identify harder conditions, but they are **heuristics, not calibrated confidence probabilities** or guarantees for individual predictions.

Zone	Low-risk MAE	High-risk MAE	High / low
ZONE_0	1.51	5.89	3.9×
ZONE_1	2.80	8.32	3.0×
ZONE_2	2.03	5.68	2.8×
ZONE_3	2.22	7.20	3.2×

Table 4: Observed group MAE in €/MWh on January–August development folds. These reused folds provide development evidence, not an independent final test of the indicator.

AI agents and LLM usage

We worked with LLM-based coding assistants throughout the project, especially when coding. It was an iterative process: we tried an approach, looked at the results, and used the assistants to help identify errors and work through possible solutions. We also discussed our EDA findings with them to explore features and model changes worth testing. We checked suggestions through tests and backtesting, keeping changes supported by the results. Finally, we used them to help organise and revise the repository, report and presentation through several rounds of feedback.

The papers, libraries and challenge materials we consulted are listed in `docs/REFERENCES.md` in the repository, including approaches tested but not retained.